

lab #7

Problema #2

$$1) S \rightarrow 0AO \mid 1B1 \mid BB$$

$$A \rightarrow C$$

$$B \rightarrow S \mid A$$

$$C \rightarrow S \mid \epsilon$$

$$a) N_0 = \emptyset$$

$$N_1 = \{C\} \quad (C \rightarrow \epsilon)$$

$$N_2 = \{C, A\} \quad (A \rightarrow C)$$

$$N_3 = \{C, A, B\} \quad (B \rightarrow A)$$

$$N_4 = \{C, A, B, S\} \quad (S \rightarrow BB)$$

$$N_5 = N_4$$

$$N = \{S, A, B, C\}$$

Producción	combinaciones
$S \rightarrow 0AO$	0AC, 00
$S \rightarrow 1B1$	1B1, 11
$S \rightarrow BB$	BB, B, B, ϵ
$A \rightarrow C$	C, ϵ
$B \rightarrow S$	S, ϵ
$C \rightarrow S$	S, ϵ
$C \rightarrow \epsilon$	se elimina

$$S \rightarrow 0AO \mid 00 \mid 1B1 \mid 11 \mid BB \mid B$$

$$A \rightarrow C$$

$$B \rightarrow S \mid A$$

$$C \rightarrow S$$

$$b) S \rightarrow B, A \rightarrow C, B \rightarrow S, B \rightarrow A, C \rightarrow S$$

$$S \rightarrow B \rightarrow A \rightarrow C \rightarrow S$$

$$U(S) = U(A) = U(B) = U(C) = \{S, A, B, C\}$$

$$\{0A0, 00, 1B1, 11, BB\}$$

$$S \rightarrow 0A0 | 00 | 1B1 | 11 | BB$$

$$A \rightarrow 0A0 | 00 | 1B1 | 11 | BB$$

$$B \rightarrow 0A0 | 00 | 1B1 | 11 | BB$$

$$C \rightarrow 0A0 | 00 | 1B1 | 11 | BB$$

$$c) S \rightarrow 00, A \rightarrow 00, B \rightarrow 00, C \rightarrow 00$$

$$P_0 = \emptyset$$

$$P_1 = \{S, A, B, C\}$$

$$P_2 = P_1$$

No terminales

$$R_0 = \{S\}$$

$$R_1 = \{S, A, B\}$$

$$R_2 = R_1$$

$$C \notin R_1$$

$$S \rightarrow 0A0 | 00 | 1B1 | 11 | BB$$

$$A \rightarrow 0A0 | 00 | 1B1 | 11 | BB$$

$$B \rightarrow 0A0 | 00 | 1B1 | 11 | BB$$

$$d) \quad X \rightarrow 0, \quad Y \rightarrow 1$$

$$S \rightarrow XAX | XX | YBY | YY | BB$$

$$A \rightarrow XAX | XX | YBY | YY | BB$$

$$B \rightarrow XAX | XX | YBY | YY | BB$$

$$D \rightarrow AX, \quad E \rightarrow BY$$

$$XAB \rightarrow XD, \quad YBY \rightarrow YE$$

$$S \rightarrow XD | XX | YE | YY | BB | S$$

$$A \rightarrow XD | XX | YE | YY | BB$$

$$B \rightarrow XD | XX | YE | YY | BB$$

$$D \rightarrow AX$$

$$E \rightarrow BY$$

$$X \rightarrow 0$$

$$Y \rightarrow 1$$

$$2) \quad S \rightarrow aAa | bBb | \epsilon$$

$$A \rightarrow C | a$$

$$B \rightarrow C | b$$

$$C \rightarrow CDE | \epsilon$$

$$D \rightarrow A | B | ab$$

$$a) \quad N_0 = \emptyset$$

$$N_1 = \{S, C\} \quad (S \rightarrow \epsilon, C \rightarrow \epsilon)$$

$$N_2 = \{S, C, A, B\} \quad (A \rightarrow C, B \rightarrow C)$$

$$N_3 = \{S, C, A, B, D\} \quad (D \rightarrow A | B)$$

$$N_4 = N_3$$

$$N = \{S, A, B, C, D\}$$

Producción	combinaciones
$S \rightarrow aAa$	aAa, aa
$S \rightarrow bBb$	bBb, bb
$A \rightarrow C$	C, ϵ
$B \rightarrow C$	C, ϵ
$C \rightarrow CDE$	CDE, CE, DE, E
$D \rightarrow A$	A, ϵ
$D \rightarrow B$	B, ϵ
$S \rightarrow E, C \rightarrow E$	se elimina

$S \rightarrow aAa \mid aa \mid bBb \mid bb$

$A \rightarrow C \mid a$

$B \rightarrow C \mid b$

$C \rightarrow CDE \mid CE \mid DE \mid E$

$D \rightarrow A \mid B \mid ab$

b) $A \rightarrow C, B \rightarrow C, C \rightarrow E, D \rightarrow A, D \rightarrow B$

$$U(S) = \{S\}$$

$$U(A) = \{A, C, E\}$$

$$U(B) = \{B, C, E\}$$

$$U(C) = \{C, E\}$$

$$U(D) = \{D, A, B, C, E\}$$

$$U(E) = \{E\}$$

$$S \rightarrow aAa \mid aa \mid bBb \mid bb$$

$$A \rightarrow a \mid CDE \mid CE \mid DE$$

$$B \rightarrow b \mid CDE \mid CE \mid DE$$

$$C \rightarrow CDE \mid CE \mid DE$$

$$D \rightarrow ab \mid a \mid b \mid CDE \mid CE \mid DE$$

c) $P_0 = \emptyset$

$$P_1 = \{S, A, B, D\}$$

$$P_2 = P_1$$

$$S \rightarrow aa, \quad A \rightarrow a, \quad B \rightarrow b, \quad D \rightarrow ab$$

$$S \rightarrow aAa \mid aa \mid bBb \mid bb$$

$$A \rightarrow a$$

$$B \rightarrow b$$

$$D \rightarrow ab \mid a \mid b$$

$$R_0 = \{S\}$$

$$R_1 = \{S, A, B\}$$

$$R_2 = R_1$$

$$S \rightarrow aAa \mid aa \mid bBb \mid bb$$

$$A \rightarrow a$$

$$B \rightarrow b$$

d) $A \rightarrow a, \quad B \rightarrow b$

$$aAa \rightarrow AAA$$

$$aa \rightarrow AA$$

$$bBb \rightarrow BBB$$

$$bb \rightarrow BB$$

$$S \rightarrow AAA \mid AA \mid BBB \mid BB$$

$$A \rightarrow a$$

$$B \rightarrow b$$

$$X \rightarrow AA, \quad Y \rightarrow BB$$

$$AAA \rightarrow AX, \quad BBB \rightarrow BY$$

$$S \rightarrow AX \mid AA \mid BY \mid BB \mid \epsilon$$

$$A \rightarrow a$$

$$B \rightarrow b$$

$$X \rightarrow AA$$

$$Y \rightarrow BB$$

$$3) S \rightarrow ASA \mid aB$$

$$A \rightarrow B \mid S$$

$$B \rightarrow b \mid \epsilon$$

$$a) N_0 = \emptyset$$

$$N_1 = \{B\} \quad (B \rightarrow \epsilon)$$

$$N_2 = \{B, A\} \quad (A \rightarrow B)$$

$$N_3 = N_2$$

$$N = \{A, B\}$$

Produccion	combinacion
$S \rightarrow ASA$	ASA, AS, SA, S
$S \rightarrow aB$	aB, a
$A \rightarrow B$	B, ϵ
$A \rightarrow S$	S
$B \rightarrow b$	b
$B \rightarrow \epsilon$	se elimina

$$S \rightarrow ASA \mid AS \mid SA \mid S \mid aB \mid a$$

$$A \rightarrow B \mid S$$

$$B \rightarrow b$$

$$b) S \rightarrow S, A \rightarrow B, A \rightarrow S$$

$$U(S) = \{S\}$$

$$U(A) = \{A, B, S\}$$

$$U(B) = \{B\}$$

$$S \rightarrow ASA | AS | SA | aB | a$$

$$B \rightarrow b$$

$$S \rightarrow ASA | AS | SA | aB | a$$

$$A \rightarrow ASA | AS | SA | aB | a | b$$

$$B \rightarrow b$$

$$c) S \rightarrow a, A \rightarrow a, B \rightarrow b$$

$$P_0 = \emptyset$$

$$P_1 = \{S, A, B\}$$

$$P_2 = P_1$$

$$R_0 = \{S\}$$

$$R_1 = \{S, A, B\}$$

$$R_2 = \{R_1\}$$

$$S \rightarrow ASA | AS | SA | aB | a$$

$$A \rightarrow ASA | AS | SA | aB | a | b$$

$$B \rightarrow b$$

$$d) S_0 \rightarrow ASA | AS | SA | aB | a$$

$$X \rightarrow a$$

$$aB \rightarrow XB$$

$$Y \rightarrow SA$$

$$ASA \rightarrow AY$$

$S_0 \rightarrow AY|AS|SA|XB|a$

$S \rightarrow AY|AS|SA|XB|a$

$A \rightarrow AY|AS|SA|XB|a|b$

$B \rightarrow b$

$X \rightarrow a$

$Y \rightarrow SA$